

GATE

CRASH COURSE

CS/IT

Subject

Data Structure & Programming

Lecture No. 02

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Topics *to be covered*

- 1 Basic operation on stack push pop
- 2 Stack as permutation generator
- 3 expression (Infix prefix postfix)
- 4 precedence of operator, Evaluation of expression



Question

#Q Which of the following is TRUE

- A** The cost of traverse for both unordered array and ordered array is same
- B** Insert in ordered array required $O(n)$ time
- C** Insertion in unordered array required $O(n)$ time
- D** The cost of Search in ordered array is better than unordered array



Stack

1. Stack is Linear data structure
2. Stack is one ended data structure
3. Index top is maintained that points to top element of stack
4. LIFO principle
5. push pop



Stack

1

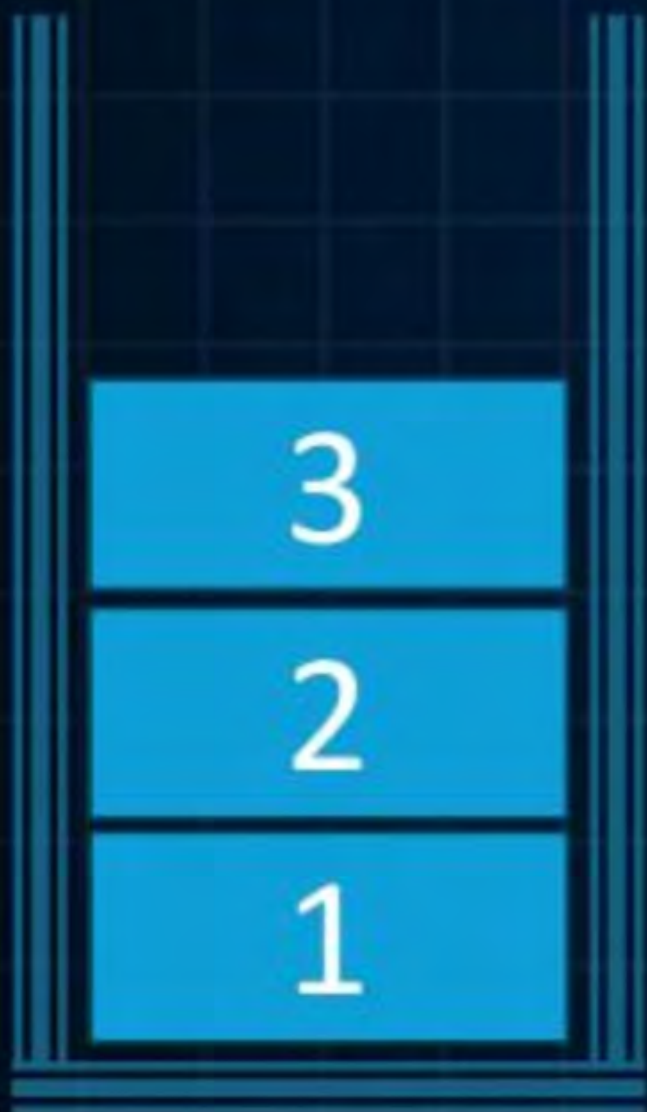


Stack



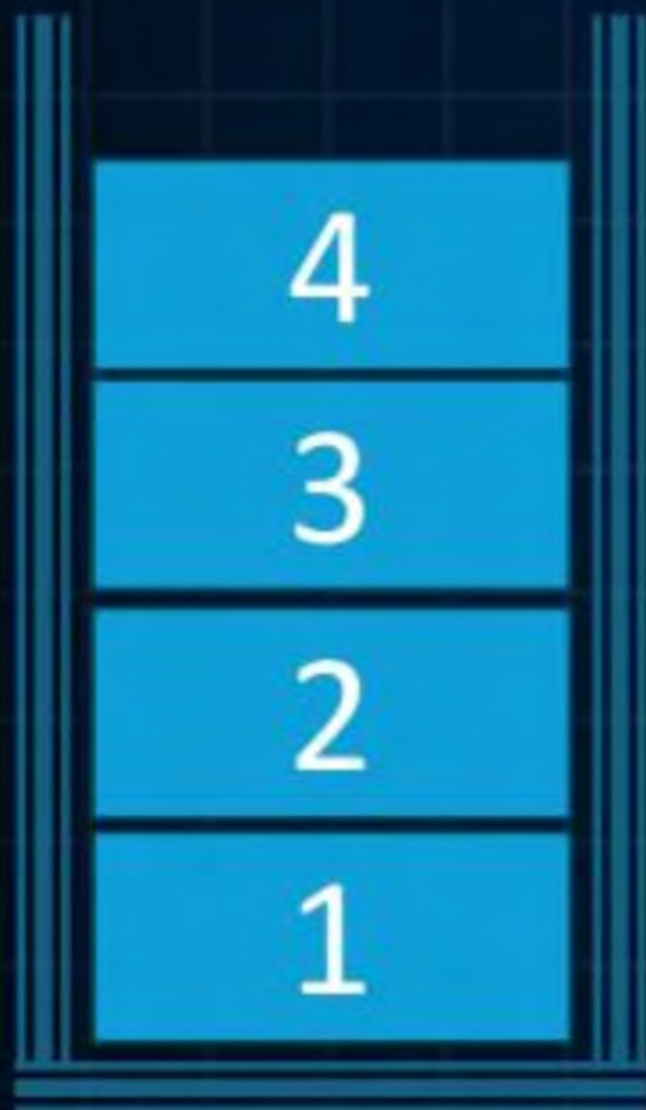


Stack



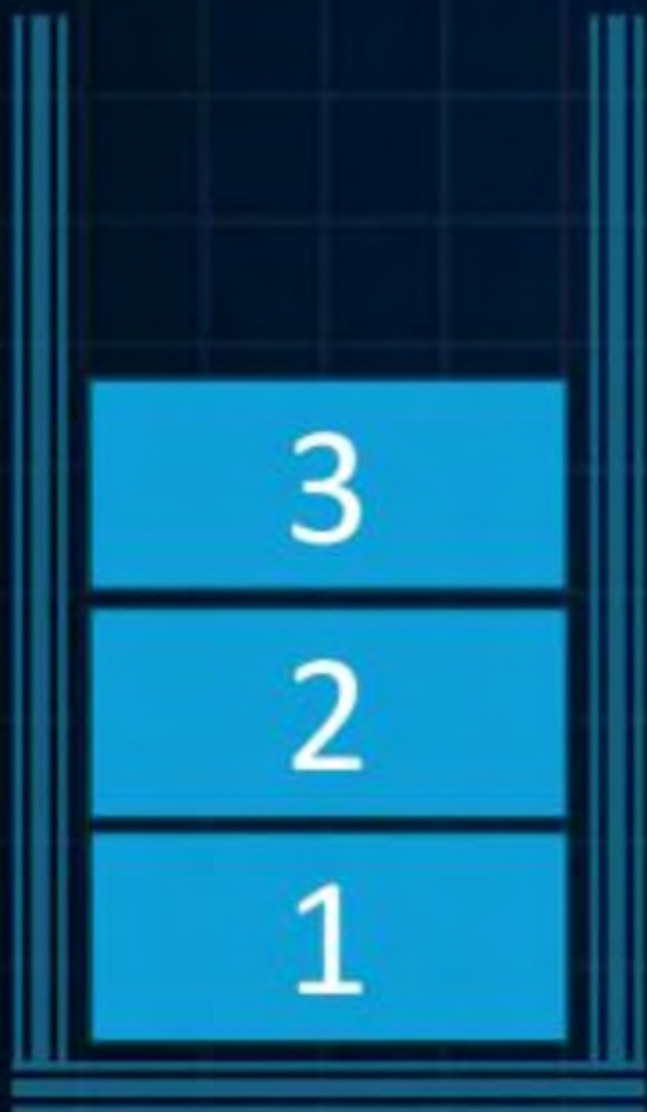


Stack





Stack





Stack





Stack

1



Stack



Question

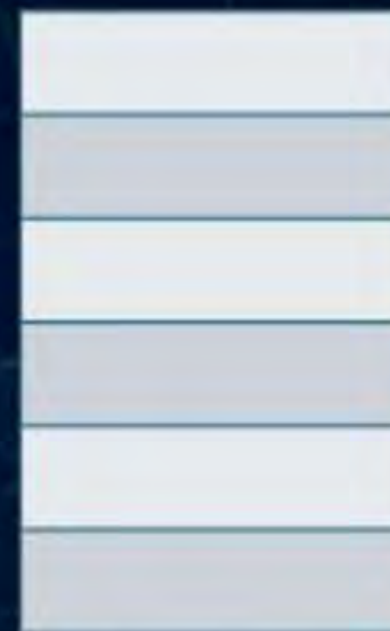
#Q The following sequence of operations is performed on a stack:

PUSH (10), PUSH (20), POP, PUSH (10), PUSH (20),
POP, POP, POP, PUSH (20), POP

The sequence of values popped out is:

- (A) 20, 10, 20, 10, 20 20, 20, 10, 10, 20
(B) ✓ 20, 20, 10, 10, 20
(C) 10, 20, 20, 10, 20
(D) 20, 20, 10, 20, 10

push → Insert on top
pop → Delete from top



Question

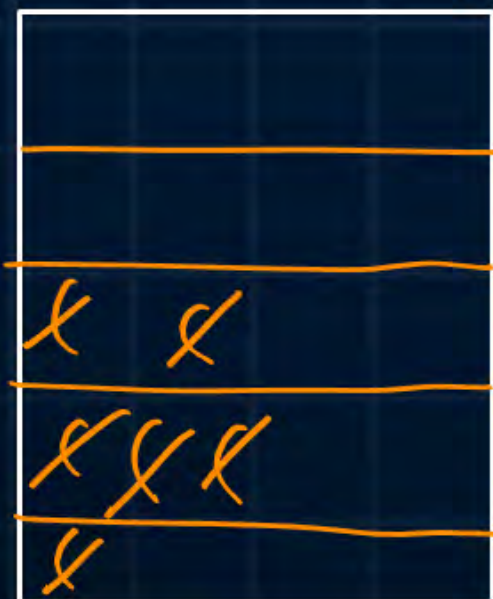
#Q Consider the usual algorithm for determining whether a sequence of parentheses is balanced. What is the maximum number of parentheses that will appear on the stack AT ANY ONE TIME when the algorithm analyzes: $(()())$?

(A) 1

(B) 2

(C) 3

(D) 4



Max. 3

PDA $\rightarrow$ FSM + stack

Balanced parentheses
(- opening push
)- pop operation



Stack as permutation generator

Consider distinct No. can be pushed in the Stack in the given order but element may pop any time. e.g. a b c

This process generates different permutation of a, b, c

Question

How many permutations are possible with a, b, c



a b c	push(a) pop push(b) pop push(c) pop ✓
a c b	push(a) pop push(b) push(c) $\begin{bmatrix} c \\ b \end{bmatrix}$ pop pop
b a c	push(a) push(b) $\begin{bmatrix} b \\ a \end{bmatrix}$ pop pop push(c) pop
b c a	push(a) push(b) pop push(c) $\begin{bmatrix} c \\ a \end{bmatrix}$ pop pop
c a b ✗	push(a) push(b) push(c) $\begin{bmatrix} c \\ b \\ a \end{bmatrix}$ c b a Not possible
c b a ✓	

Accuracy

+
Speed

With 3 elements 5 permutation are possible

No. of permutation generated

$$\text{Catalan No.} = \frac{1}{n+1} {}^{2n}C_n \quad \leftarrow \begin{array}{l} \text{Counting} \\ \text{unlabeled trees} \end{array}$$

$$C_1 = 1$$

$$C_3 = 5$$

$$C_5 = 42$$

$$C_2 = 2$$

$$C_4 = 14$$

$$C_6 = 132$$

+ counting
BST



Expression

Expression Infix: $a + b$

← Evaluation Required
multiple Scans

Single
Scan

prefix: $+ab$

postfix: $ab+$

precedence
& Associativity
already considered



Precedence & Associativity

Unary minus	-	$\frac{-5}{\uparrow}$ unary minus	Highest
Exponentiation	$\uparrow$	Right to Left	$(2 \uparrow 4 \uparrow 3)$
Multiplicative	$* / \%$	Left to Right	
Additive	$+ -$	Left to Right	
Equal	$=$	Right to Left	Lowest $a = (b = c)$



Expression

$$A - B/C * D \uparrow E$$

Diagram showing the expression $A - B/C * D \uparrow E$ with annotations: 4 under A, 2 under the minus sign, 3 above B/C, and 1 above the arrow in $D \uparrow E$.

operand $\rightarrow$
variable Name
Numeric

$$\text{prefix} = A - B/C * \uparrow DE$$

operand

$$= A - /BC * \uparrow DE$$

operand operand

$$= A - * /BC \uparrow DE$$

$$= - A * /BC \uparrow DE$$

a+b

ab+

$$A - B/C * \underline{DE \uparrow}$$

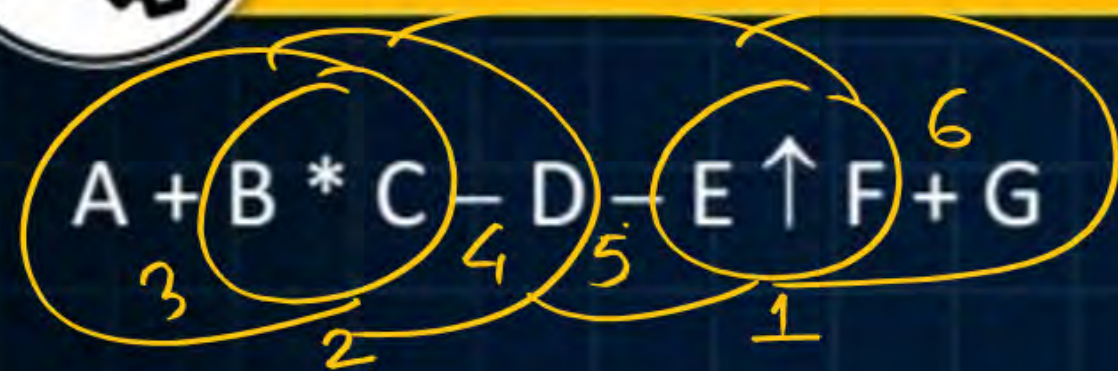
$$A - \underline{BC /} * \underline{DE \uparrow}$$

$$A - \underline{BC / DE \uparrow} *$$

$$A \ BC / DE \uparrow * -$$



Expression



$$= A + \underline{*BC} - D - \underline{\uparrow EF} + G$$

$$= + - - + A * B C D \uparrow E F G$$

postfix

$$A + \underline{BC*} - D - \underline{EF\uparrow} + G$$

$$= \underline{ABC*+} - D - EF\uparrow + G$$

$$= \underline{ABC*+D-} - EF\uparrow + G$$

$$= ABC*+D-EF\uparrow-+G$$

$$= ABC*+D-EF\uparrow G+$$

Question

$$f \uparrow g \uparrow h = f \uparrow \underline{g \uparrow h} = \uparrow f \uparrow g \uparrow h$$

$$c * d / e = \underline{*cd} / e = \underline{/ * cde}$$

The **prefix** expression for the infix expression

$$a = -b + \boxed{c * d / e} + \boxed{f \uparrow g \uparrow h} - i * j$$

- ~~(a)~~ = a - + + - b / * cde $\uparrow$ f $\uparrow$ gh * ij a = - b + / * cde + $\uparrow$ f $\uparrow$ gh - * ij
 (b) = a - + + - / * bcde $\uparrow$ f $\uparrow$ gh * ij a = - + + - b / * cde $\uparrow$ f $\uparrow$ gh * ij
 (c) = a + - + - b / * cde $\uparrow$ f $\uparrow$ gh * ij
 (d) = a - + - - b / * cde $\uparrow$ f $\uparrow$ gh * ij a = - + + - b / * cde $\uparrow$ f $\uparrow$ gh * ij

[A]

Question

The attribute of three arithmetic operators in some programming language are given below.

PYQ : 2016

OPERATOR	PRECEDENCE	ASSOCIATIVITY	ARITY
+	High	Left	Binary
-	Medium	Right	Binary
*	Low	Left	Binary

The value of the expression $2 - 5 + 1 - 7 * 3$ in this language is 9.

$$\begin{aligned} & 2 - 5 + 1 - 7 * 3 \\ & 2 - (5 - 1) * 3 \\ & 2 - (4) * 3 \\ & 2 - 12 \\ & -10 \end{aligned}$$
$$\begin{aligned} & 2 - (-1) * 3 \\ & 3 * 3 = 9 \end{aligned}$$



Expression

Consider the operator precedence and associativity rules for the *integer* arithmetic operators given in the table below.

Operator	Precedence	Associativity
+	Highest	Left
-	High	Right
*	Medium	Right
/	Low	Right

$$3 + 1 + 5 * 2 / 7 + 2 - 4 - 7 - 6 / 2$$

$$9 * 2 / 9 - 4 - 7 - 6 / 2$$

$$9 * 2 / 9 - 4 - 1 / 2$$

$$9 * 2 / 9 - 3 / 2$$

$$9 * 2 / 6 / 2$$

The value of the expression $3 + 1 + 5 * 2 / 7 + 2 - 4 - 7 - 6 / 2$ as per the above rules is [6].

$$18 / (6 / 2) \\ 18 / 3 = 6 \\ \uparrow$$



Topic: Evaluation of Postfix Expression

6, 2, 3

6 2 3 + - 3 8 2 / + * 2 ↑ 3 + ↑ 3 + Single scan

			2									
3			8	$8/2=4$								
2	$2+3=5$		3	3	7		2		3			
6	6	$6-5=1$	1	1	1	$1*7=7$	7	$7^2=49$	49	$49+3=52$		

operator :

operand₂ = pop()

operand₁ = pop()

value = operand₁ op operand₂

push(value)



Topic: Evaluation of Postfix Expression

$$ab+ = (a+b)$$

6 2 3 + - 3 8 2 / + * 2 ↑ 3 +

$$6, 2, 3 +$$

$$6, (2+3)$$

$$\underline{6, 5} -$$

$$1, 3, 8, 2 /$$

$$1, 3, 4 +$$

$$1, 7 *$$

$$7, 2 \uparrow$$

$$7 \uparrow 2 = 49$$

$$49, 3 + = \underline{52}$$



Topic: Evaluation of Postfix Expression

23 * 422, *, /, +, 35, +, -

2, 3 *

Answer is -1.

6, 4, 2, 2 *

6, 4, 4 /

6, 1 +

7, 3, 5 +

7, 8 -

(7 - 8) = -1

Question

$$\underline{108} - 54 = \underline{54}$$

Q Consider the following expression: $(6\ 3\ *\ 4\ 2\ +\ *) - (+\ *\ 4\ 3\ *\ +\ 2\ 5\ *\ 6\ 1)$
Each sub-expression is either in postfix or prefix notation. All operands are single-digit numbers. What will be the result after evaluating the expression?

$$\begin{aligned} 6\ 3\ * \\ 18\ 4\ 2\ + \\ 18\ 6\ * \\ 108 \end{aligned}$$

$$\begin{aligned} & \quad \quad \quad \underline{* 6\ 1} \\ & \quad \quad \quad + 2\ 5\ 6 \\ & \quad \quad \quad * 7\ 6 \\ & \quad \quad \quad * 4\ 3\ 4\ 2 \\ & \quad \quad \quad + 12\ 4\ 2 \\ & \quad \quad \quad = 54 \end{aligned}$$

Question

Monday: 9.30 pm

#Q.

The following postfix expression is evaluated using a stack:

$$9\ 3\ +\ 4\ 4\ *\ X\ /\ 1\ ^\ -\ X\ +$$

If the final value obtained after evaluation of the above postfix expression is 18, then the sum of all possible values of X is [6].

$$9, 3 +$$

$$12, 4, 4 *$$

$$12, 16, X, /$$

$$12, \frac{16}{X}, 1 ^$$

$$12, \frac{16}{X} -$$

$$12 - \frac{16}{X}, X +$$

$$12 - \frac{16}{X} + X = 18$$

$$12X - 16 + X^2 = 18X$$

$$X^2 - 6X - 16 = 0$$

$$\boxed{\frac{-b}{a} = \frac{-(-6)}{1} = 6}$$

Sum of

$$X^2 - 8X + 2X - 16$$

$$(X-8)(X+2) = 0$$

$$X = 8$$

$$X = -2$$

prefix

Right to Left

/ * + 2 2 3 * 4 3

prefix evaluation

+ a b

$\Rightarrow$ a * b

$\downarrow$
* 4 3 $\Rightarrow$ 4 * 3

+ 2, 2, 3, 12 $\Rightarrow$ (2 + 2)

* 4 3, 12 $\Rightarrow$ (4 * 3) = 12

/ 12, 12 $\Rightarrow$ 12 / 12 = 1

①

Monday 9.30 pm ←

Thank
THANK



Keep Hustling!